

Transport and Telecommunication Institute &
UWE Bristol

Machine Learning

Time-Series Regression for Predicting Rack Inlet Temperature in a Free-Cooled Data Center

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1. INTRODUCTION

1.1. Background and Problem Definition

The increasing demand for digital services has led to rapid growth in data center infrastructure, making energy efficiency and thermal management critical operational challenges. Inefficient thermal control can result in increased operational costs, reduced system performance, and a higher risk of hardware failures.

Free-cooled data centers use outside air as the primary cooling mechanism, offering significant potential for energy savings compared to traditional cooling approaches. However, this design also introduces greater sensitivity to environmental conditions such as outside temperature, humidity, airflow, and pressure. As a result, maintaining stable rack inlet temperature becomes more challenging. Since rack inlet temperature is a key indicator of thermal safety, accurate prediction of its future values can support proactive cooling management.

1.2. Research Aim and Objectives

The aim of this study is to develop and evaluate supervised time-series regression models for predicting rack inlet temperature in a free-cooled data center using environmental and energy sensor data.

Research Objectives

1. Analyse the characteristics of the data center sensor dataset;
2. Preprocess and engineer features from multivariate time-series data;
3. Implement and compare selected supervised regression models for rack inlet temperature prediction;
4. Evaluate model performance using appropriate regression-based metrics;
5. Assess the suitability and limitations of the applied models in the context of data center cooling.

2. DATASET DESCRIPTION AND EXPLORATORY DATA ANALYSIS

2.1. Dataset Overview

This study uses the **An Air Free-Cooled Tropical Data Center (TDC1.0) Dataset**, provided by **Le, Van Duc (2021)**. The dataset contains high-resolution sensor measurements collected from an air free-cooled data center testbed operating in a tropical climate during 2018–2019.

The experiments are organised into folders corresponding to **controlled supply air temperature setpoints**, ranging from **25°C to 37°C**, where each setpoint represents an independent, steady-state operating regime. For every setpoint, **22 sensor files** are available, covering thermal conditions (rack and room temperatures), relative humidity, pressure differences, airflow rates, outdoor environmental variables, and energy consumption.

To increase data volume and reduce the impact of small-sample bias, data from **all controlled setpoints (25–37°C)** were combined. All variables were resampled to a **30-minute resolution** to ensure temporal alignment and reduce short-term noise. The final dataset consists of **780 observations**, with no missing values across the selected variables (`wide_30min_controlled_all.csv`).

	setpoint	files	min_rows	max_rows	avg_missing
0	25	22	1800	21600	0.0
1	26	22	1800	21600	0.0
2	27	22	1800	21600	0.0
3	28	22	1800	21600	0.0
4	29	22	1800	21600	0.0
5	30	22	1800	21600	0.0
6	31	22	1800	21600	0.0
7	32	22	1800	21600	0.0
8	33	22	1800	21600	0.0
9	34	22	1800	21600	0.0
10	35	22	1800	21600	0.0
11	36	22	1800	21600	0.0
12	37	22	1800	21600	0.0

Figure 1. Dataset structure and coverage

2.2. Exploratory Data Analysis

Exploratory data analysis confirms that the dataset is clean, stable, and suitable for supervised learning. The **average cold rack temperature** was selected as the target variable, as it directly reflects cooling effectiveness and is consistently measured across all experimental regimes.

Descriptive statistics show a clear separation between temperature setpoints, with low variance of cold rack temperature within each regime and a near-linear increase across regimes. In contrast, variables such as rack pressure difference, airflow rate, and relative humidity exhibit wider distributions, indicating active control dynamics within the cooling system.

Figure 2 illustrates descriptive statistics which show consistent coverage across controlled supply temperature setpoints (25–37 °C) with no missing values. Outdoor conditions exhibit low variability, while airflow rate and rack pressure difference display wider ranges, reflecting operational adjustments under different thermal regimes. Rack temperatures increase monotonically with higher setpoints, confirming physically consistent and reliable measurements.

	count	mean	std	min	25%	50%	75%	max
supply_setpoint_label	780.0	31.000000	3.744058	25.000000	28.000000	31.000000	34.000000	37.000000
supply_temp_setpoint_meas	780.0	31.000000	3.744058	25.000000	28.000000	31.000000	34.000000	37.000000
outdoor_temp	780.0	24.235006	0.784419	21.519700	23.708150	24.265450	24.810342	26.500900
outdoor_rh	780.0	24.235006	0.784419	21.519700	23.708150	24.265450	24.810342	26.500900
airflow_rate	780.0	7615.384615	4138.118520	2500.000000	5000.000000	7500.000000	10000.000000	50000.000000
dp_rack	780.0	17.898657	10.233984	3.490064	8.915565	16.375117	25.393100	61.938869
cold_rack_temp_avg	780.0	31.068184	3.717191	24.753143	27.814542	31.255113	34.309610	37.585760
hot_rack_temp_avg	780.0	33.080082	3.495148	27.189024	30.123850	33.045930	36.092005	39.754725
delta_t	780.0	2.011898	0.787357	0.022422	1.961505	2.169502	2.458459	3.533429
cold_rack_rh_avg	780.0	61.455290	12.745985	40.037615	50.535315	60.726862	72.888353	85.328100
hot_rack_rh_avg	780.0	57.196650	11.993590	37.162727	47.032624	56.309092	67.579845	81.590812

Figure 2. Descriptive statistics

Time-series inspection reveals stable thermal behavior within each setpoint, characterised by small short-term fluctuations and no observable long-term drift. While the absolute temperature level changes with the setpoint, intra-regime dynamics remain consistent across experiments. Visualisaions are avaible in output directory in. the project.

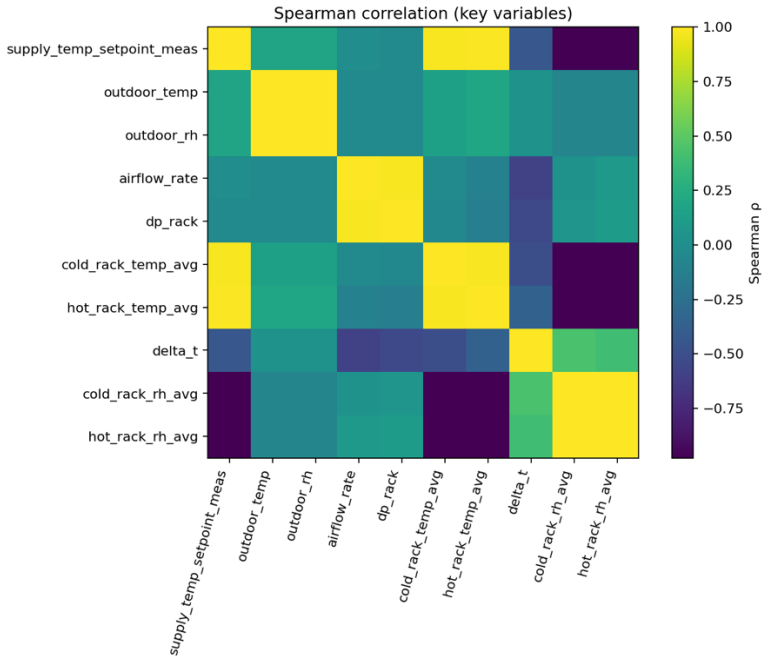


Figure 3. Spearman correlation matrix

Figure 3 presents the Spearman correlation matrix for key environmental and operational variables. The cold rack temperature shows strong monotonic dependence on the supply temperature setpoint and hot rack temperature, while moderate correlations are observed with airflow rate and rack pressure difference. Outdoor temperature and humidity exhibit weaker associations, suggesting an indirect influence under controlled operating conditions.

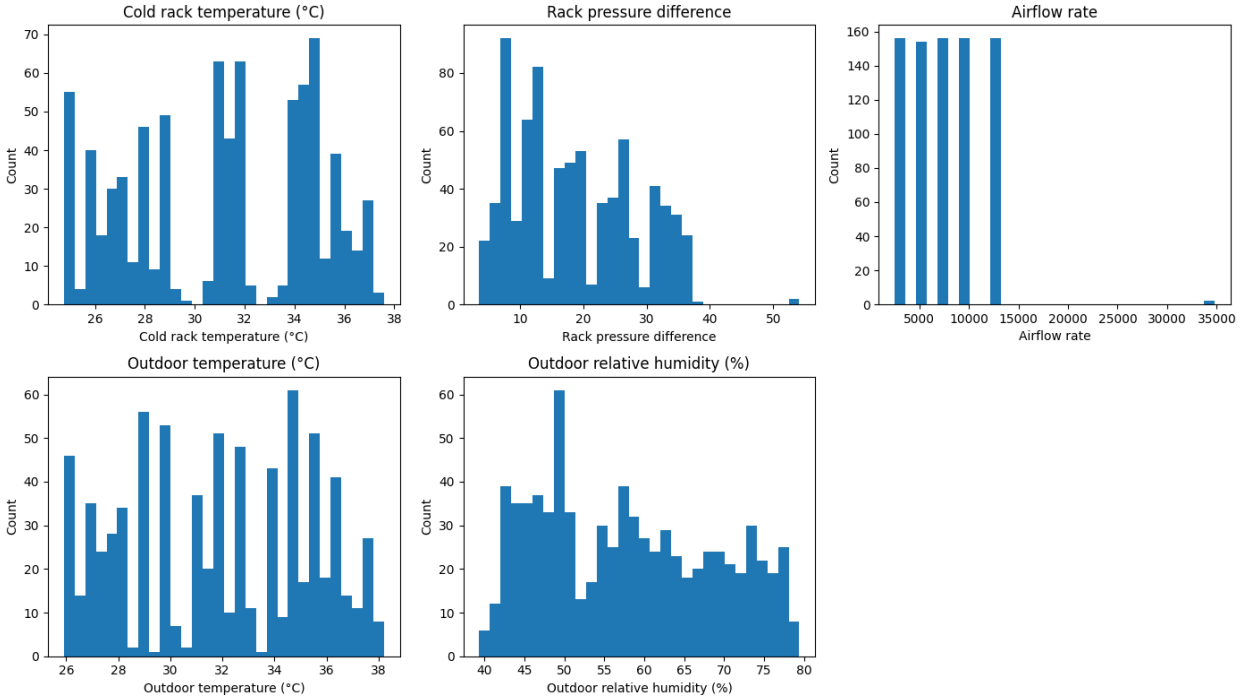


Figure 4. Selected variables distribution

Figure 4 illustrates the marginal distributions of selected variables. Cold rack temperature and outdoor temperature show clear regime-driven separation, whereas airflow rate and rack pressure difference exhibit wider, skewed distributions, reflecting adaptive control responses. Relative humidity presents broader variability, consistent with external environmental effects rather than direct thermal control.

3. DATA PREPROCESSING AND FEATURE ENGINEERING

3.1. Data Preprocessing

All experiments conducted under **controlled supply air temperature setpoints (25–37 °C)** were merged to form a single modelling dataset. This aggregation increases statistical power and ensures consistent learning across operating regimes. Raw sensor files were temporally aligned and resampled to a **30-minute resolution**, which reduces short-term noise while preserving system-level dynamics.

Redundant sensors measuring the same physical quantities were averaged (e.g., rack temperatures, humidity sensors), and only variables consistently available across all regimes were retained. Signals with incompatible semantics or sparse coverage were excluded. After preprocessing, the dataset contained **689 observations with 70 variables**, with **no missing values**. The resulting modelling table was saved as `model_df.csv`.

The target variable was defined as the **average cold rack temperature**, representing cooling effectiveness under each operating condition. **Figure 5** illustrates the temporal evolution of the target across all controlled setpoints.

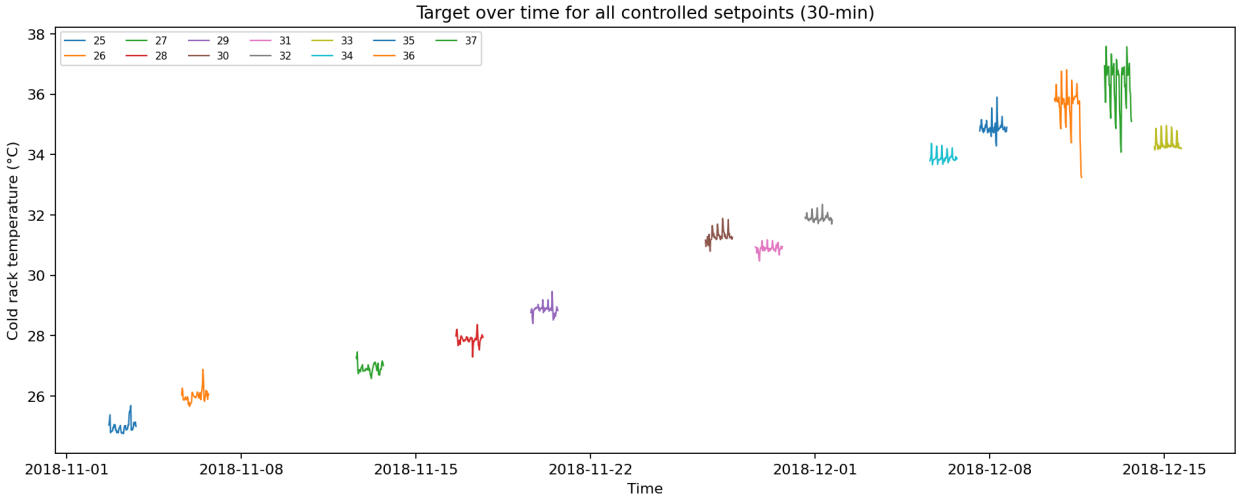


Figure 5. Target over time for all controlled setpoints

3.2. Feature Engineering

Feature engineering aimed to capture both instantaneous system states and short-term temporal dependencies. Rolling mean features were computed to smooth transient fluctuations, while lagged variables were introduced to encode delayed thermal responses. Environmental (outdoor temperature and humidity), airflow, pressure difference, and rack-level thermal variables were retained to reflect the physical cooling process. The final feature set is documented in `feature_list.csv`

Figure 6 shows the distribution of the target variable across all regimes, confirming clear separation between setpoints and stable intra-regime behavior.

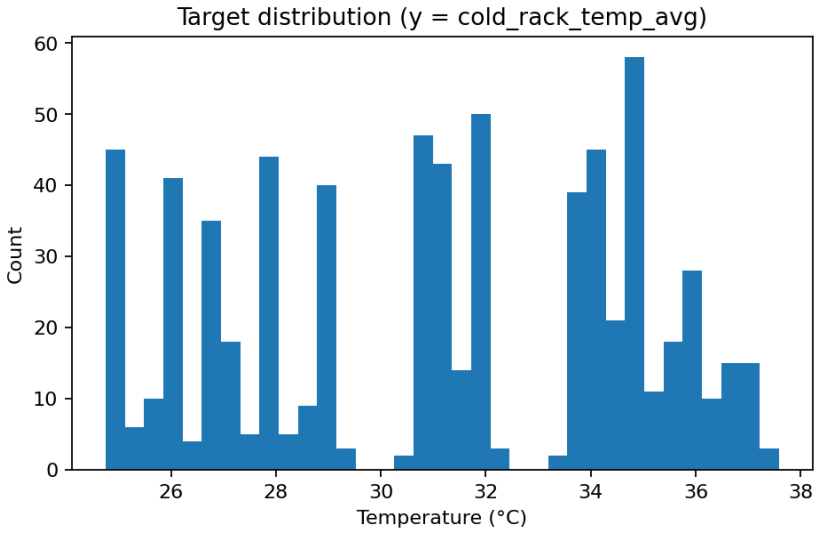


Figure 6. Target distribution

To assess monotonic relationships between predictors and the target, Spearman correlation analysis was performed. **Figure 7** highlights the strongest associations, guiding feature relevance without assuming linearity.

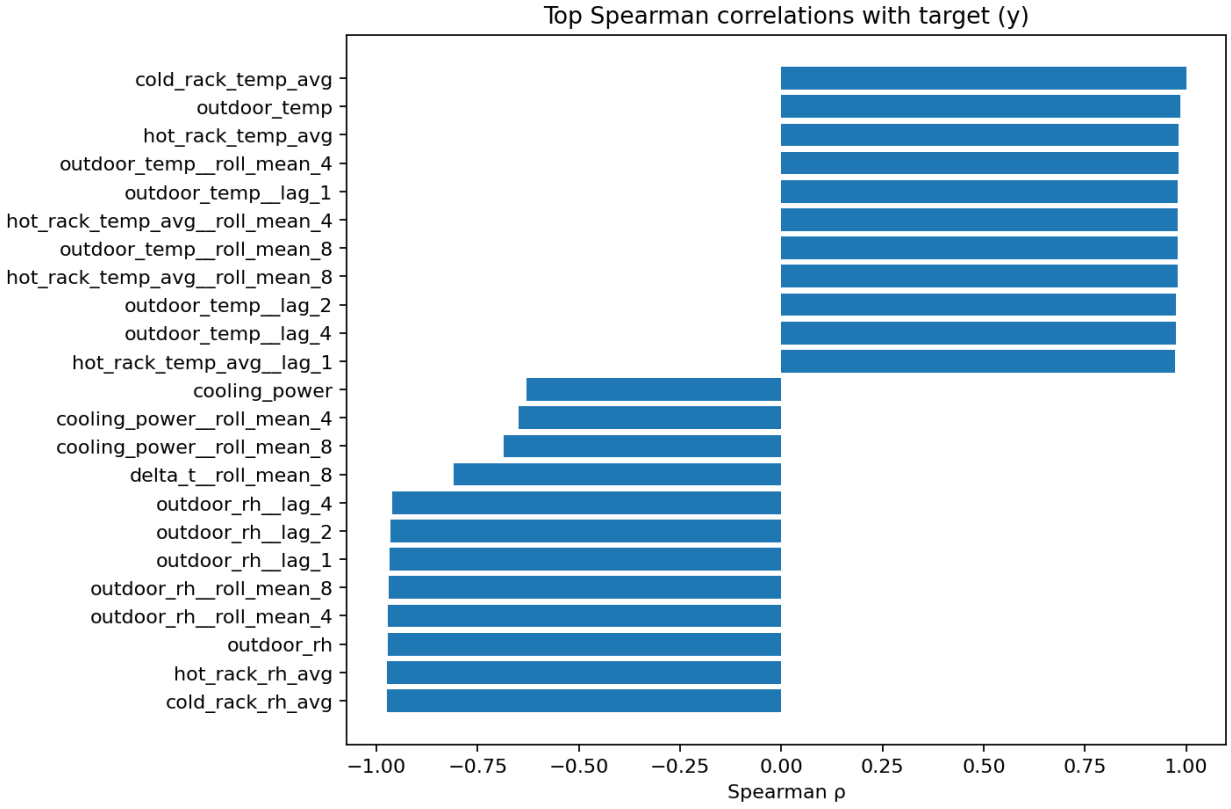


Figure 7. Top Spearman correlations with the target

The engineered feature set balances interpretability and predictive capacity and serves as the direct input for model training in Section 4.

All data preprocessing, feature engineering, model training, evaluation scripts, and generated figures are fully reproducible. The complete project codebase, including intermediate datasets and plotting scripts, is publicly available at: **https://github.com/anastasijazu/ml_project_2025**

4. PROBLEM FORMULATION AND METHODOLOGY

4.1. Learning Problem Definition

The objective of this study is to model the thermal behaviour of an air free-cooled data center by predicting the **average cold rack temperature (**y=cold_rack_temp_avg) using supervised regression.

The input space consists of environmental, operational, and engineered temporal features derived from sensor measurements, aggregated to a **30-minute resolution**. All experiments conducted under **controlled supply air temperature setpoints (25–37 °C)** are combined into a single dataset to improve statistical robustness and cross-regime generalisation.

Formally, the task is to learn a function f: $\mathbf{X}_t \rightarrow y_t$ where $\mathbf{X}_t$ represents the feature vector at time t. A **time-aware train-test split** is applied to prevent information leakage, ensuring that predictions are made strictly on future observations.

4.2. Selected Regression Models

Three regression approaches of increasing modelling capacity are evaluated:

- **Baseline (Setpoint Mean):**

  Predicts the mean cold rack temperature observed during training for each supply temperature regime. This serves as a lower-bound reference.

- **Linear Regression:**

  A multivariate linear model trained on standardized features. It provides a transparent benchmark for assessing linear dependencies between predictors and the target.

- **Ridge Regression:**

  An L2-regularised extension of linear regression used to stabilize coefficient estimates under multicollinearity while retaining interpretability.

- **Random Forest Regressor:**

  A non-linear ensemble model based on decision trees, included to capture potential non-linear interactions and higher-order effects. Feature importance analysis is performed for model interpretability.

4.3. Model Implementation Details

All models are implemented using **scikit-learn**. Linear and Ridge regressions are embedded in a pipeline with **standardisation (z-score scaling)** applied to all input features. Random Forest models are trained on unscaled features.

The dataset used for modelling is stored as `output/section3/model_df.csv`.

A **chronological split** assigns the first 75 % of samples to training and the remaining 25 % to testing. No cross-validation is applied, as preserving temporal structure is prioritised over random resampling.

For Random Forest, the final configuration uses 300 trees, maximum depth of 12, fixed random seed for reproducibility.

5. MODEL TRAINING AND EVALUATION

5.1. Training Procedure

Models are trained on **516 observations** and evaluated on **173 unseen observations**, covering all controlled setpoints. Training is performed once per model without hyperparameter tuning, as the goal is comparative analysis rather than optimization.

The Random Forest model is trained with 300 trees and bounded depth to limit overfitting. Feature importance scores are extracted post-training to support interpretability.

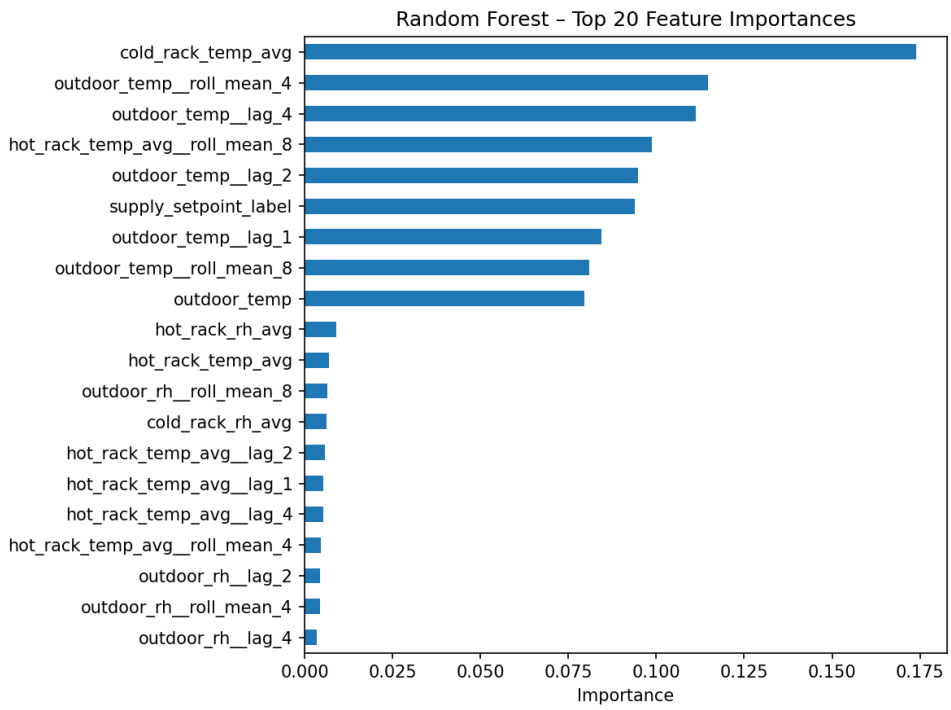


Figure 8. RF - Top 20 Feature Importance

5.2. Evaluation Metrics and Results

Results, reported in *model_results_overall.csv*, demonstrate clear differences between the models. **Linear Regression** achieves an almost perfect fit on the test set (MAE $\approx 10^{-15}$, RMSE $\approx 10^{-15}$, $R^2 = 1.00$), as illustrated by the near-exact alignment with the identity line in **Figure 9** and the complete overlap between predicted and observed trajectories in **Figure 10** (linear_time_actual_vs_pred.png). However, this apparent perfection should not be interpreted as a physically meaningful predictive capability. Instead, it reflects the highly structured and deterministic nature of the controlled experimental setup, where the target variable is strongly governed by the supply temperature setpoint and linearly related operational variables. Consequently, the linear model effectively interpolates between regimes rather than learning complex dynamics, and its performance likely overestimates generalisation under less controlled conditions.

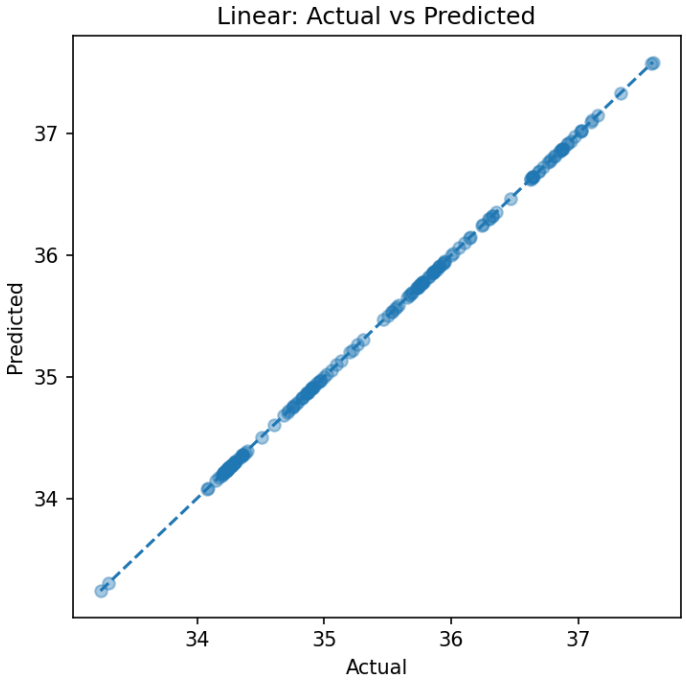


Figure 9. Linear Scatter Plot

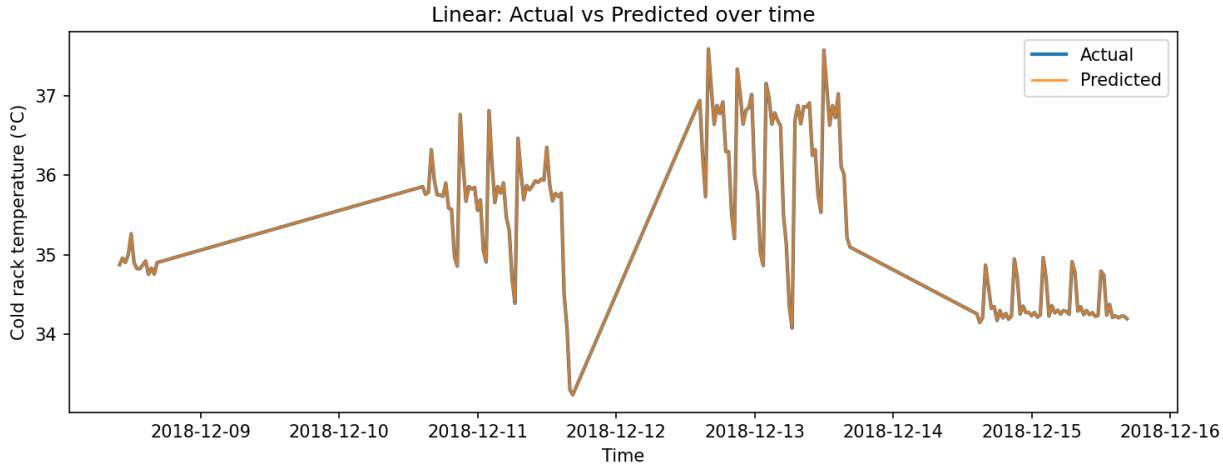


Figure 10. Linear time prediction

The **baseline model**, which predicts the mean cold rack temperature per supply setpoint, performs substantially worse (MAE ≈ 5.36 °C, RMSE ≈ 5.67 °C, $R^2 = -31.28$), confirming that simple regime-wise averaging fails to capture short-term intra-regime variability and transient behaviour. This establishes a meaningful lower bound for model comparison.

Ridge Regression provides a more realistic and robust alternative, achieving strong performance (MAE ≈ 0.09 °C, RMSE ≈ 0.13 °C, $R^2 \approx 0.98$). The scatter plot in **Figure 11** and the temporal comparison in **Figure 12** (show close agreement with observations while allowing small

deviations during rapid temperature transitions. The inclusion of L2 regularisation mitigates multicollinearity among correlated features and prevents the degenerate behaviour observed in the unregularised linear model, resulting in more credible generalisation.

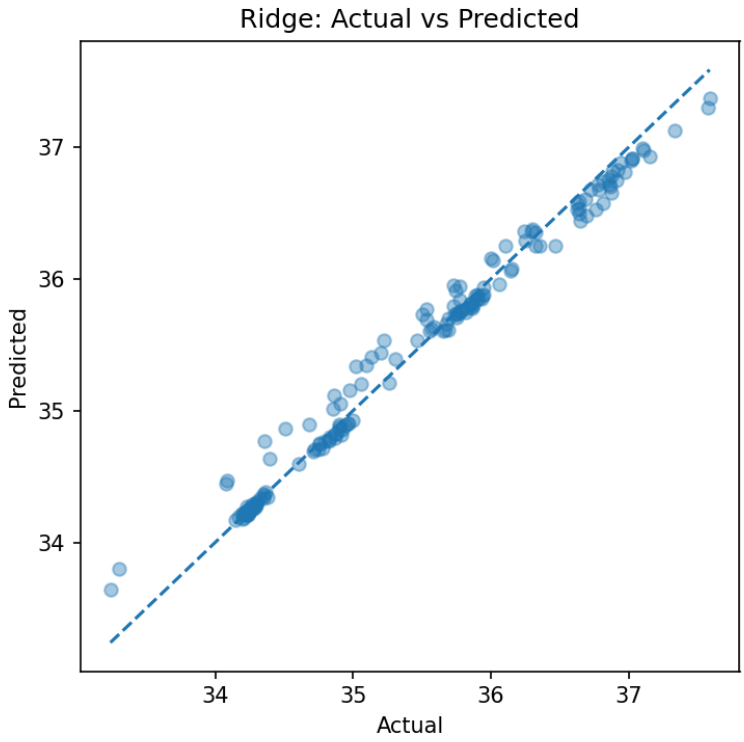


Figure 11. Ridge Scatter Plot

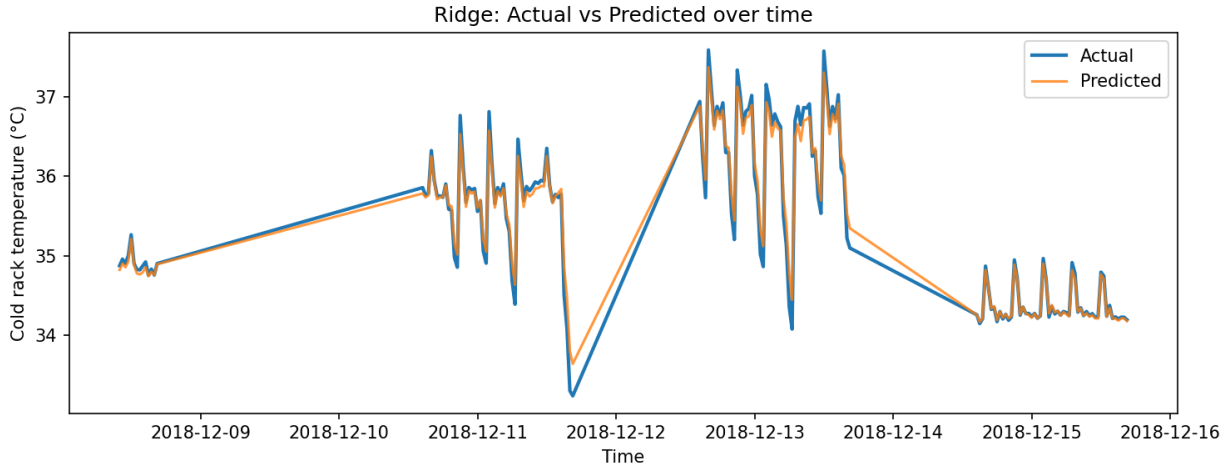


Figure 12. Ridge Time Prediction

In contrast, the **Random Forest** model underperforms relative to linear approaches (MAE ≈ 0.76 °C, RMSE ≈ 0.91 °C, R^2 ≈ 0.17). As shown in **Figures 13 and 14**, the model smooths abrupt

temperature changes and systematically underestimates regime-level shifts. This behavior indicates that tree-based averaging struggles to represent the sharp, regime-driven structure of the data, despite its ability to model non-linear interactions. Per-regime evaluation (*model_results_per_regime.csv*) further confirms that Random Forest errors increase at higher supply temperature setpoints, whereas Ridge Regression maintains stable accuracy across all regimes.

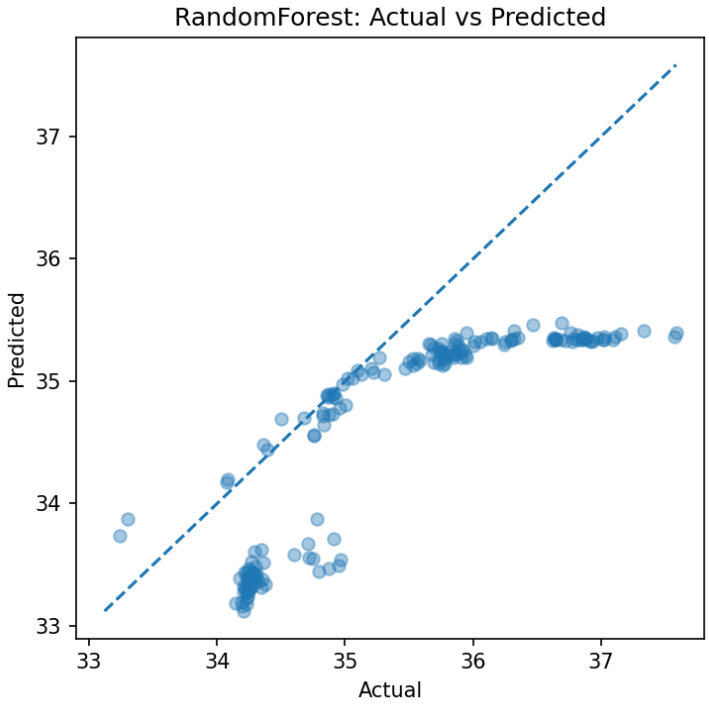


Figure 13. Random Forest Scatter

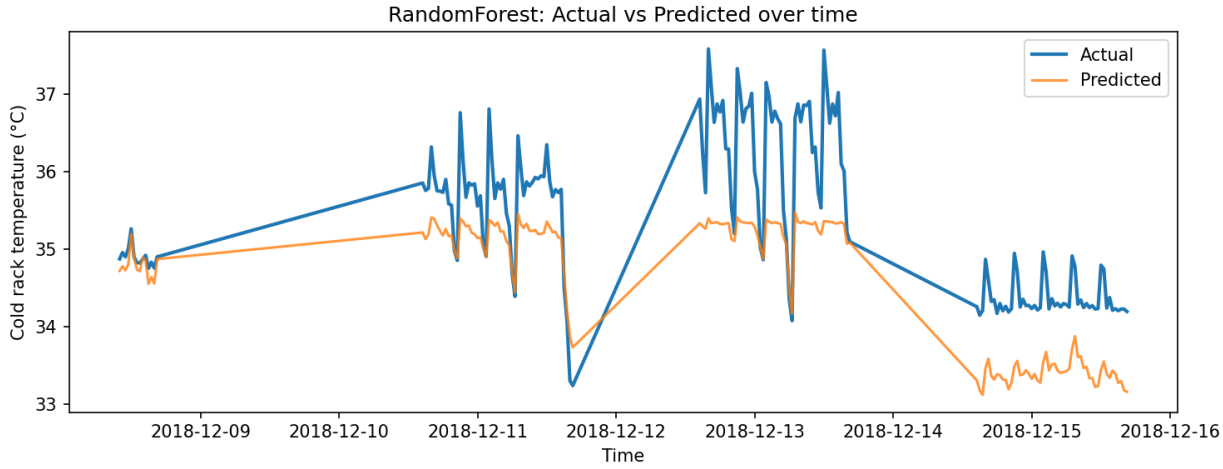


Figure 14. Random Fores Time Prediction

Overall, these results indicate that **cold rack temperature in this controlled environment is dominated by strong linear relationships**, primarily driven by the imposed supply temperature regimes. While Linear Regression achieves numerically perfect results, **Ridge Regression provides the most reliable and physically plausible model**, balancing accuracy, stability, and generalisation. Additional model complexity, as introduced by Random Forests, does not yield performance gains in this setting.

6. MODEL COMPARISON AND REJECTION ANALYSIS

6.1. Performance Comparison

Comparative analysis across all evaluated models highlights clear differences in suitability for the studied task. Linear and Ridge regression models substantially outperform the baseline and Random Forest approaches, demonstrating consistently low prediction errors across all controlled temperature regimes. While Linear Regression achieves numerically perfect performance, this behaviour reflects the highly structured and deterministic nature of the controlled experimental setup rather than superior generalisation. Ridge Regression introduces regularisation that constrains coefficient magnitude, resulting in slightly reduced accuracy but improved robustness and interpretability.

Random Forest exhibits inferior performance, particularly during regime transitions, where it smooths sharp temperature changes and underestimates setpoint-driven shifts. This confirms that increased model complexity does not translate into improved predictive capability for this dataset.

6.2. Rejected Approaches and Limitations

The baseline mean-by-setpoint model is rejected due to its inability to capture intra-regime variability and transient dynamics, resulting in poor predictive accuracy. The Random Forest model is also deemed unsuitable, as its non-linear structure fails to represent the predominantly linear relationships governing cold rack temperature under controlled conditions.

A key limitation of this study is the reliance on controlled experimental data, which constrains model evaluation to steady-state operating regimes. Consequently, the strong performance of linear models may not directly generalise to uncontrolled or highly dynamic environments. Future work should consider incorporating uncontrolled operational data and adaptive modelling strategies to assess robustness under real-world variability.

7. CONCLUSION

This study investigated the prediction of cold rack inlet temperature in an air free-cooled tropical data center using the TDC1.0 dataset under controlled supply air temperature regimes. By combining data across all controlled setpoints and applying consistent preprocessing and feature engineering at a 30-minute resolution, a robust regression framework was established.

The experimental results demonstrate that the thermal behavior of the cold rack temperature in this controlled environment is governed by strongly linear relationships. Linear and Ridge regression models achieved consistently low errors across all operating regimes, with Ridge regression preferred due to its regularization and improved robustness. Although Linear Regression achieved near-perfect predictive accuracy, this result must be interpreted cautiously, as the structured nature of the controlled regimes and strong correlation between features and target may inflate apparent performance. In contrast, the Random Forest model underperformed, smoothing sharp transitions and failing to capture regime-level structure, indicating that additional model complexity does not yield benefits for this task.

Overall, the findings highlight the importance of aligning model complexity with system physics. In controlled engineering systems with stable dynamics, simple and interpretable models can outperform more complex alternatives. Future work should extend this analysis to uncontrolled operating conditions and longer time horizons to assess generalization under higher environmental variability.

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